

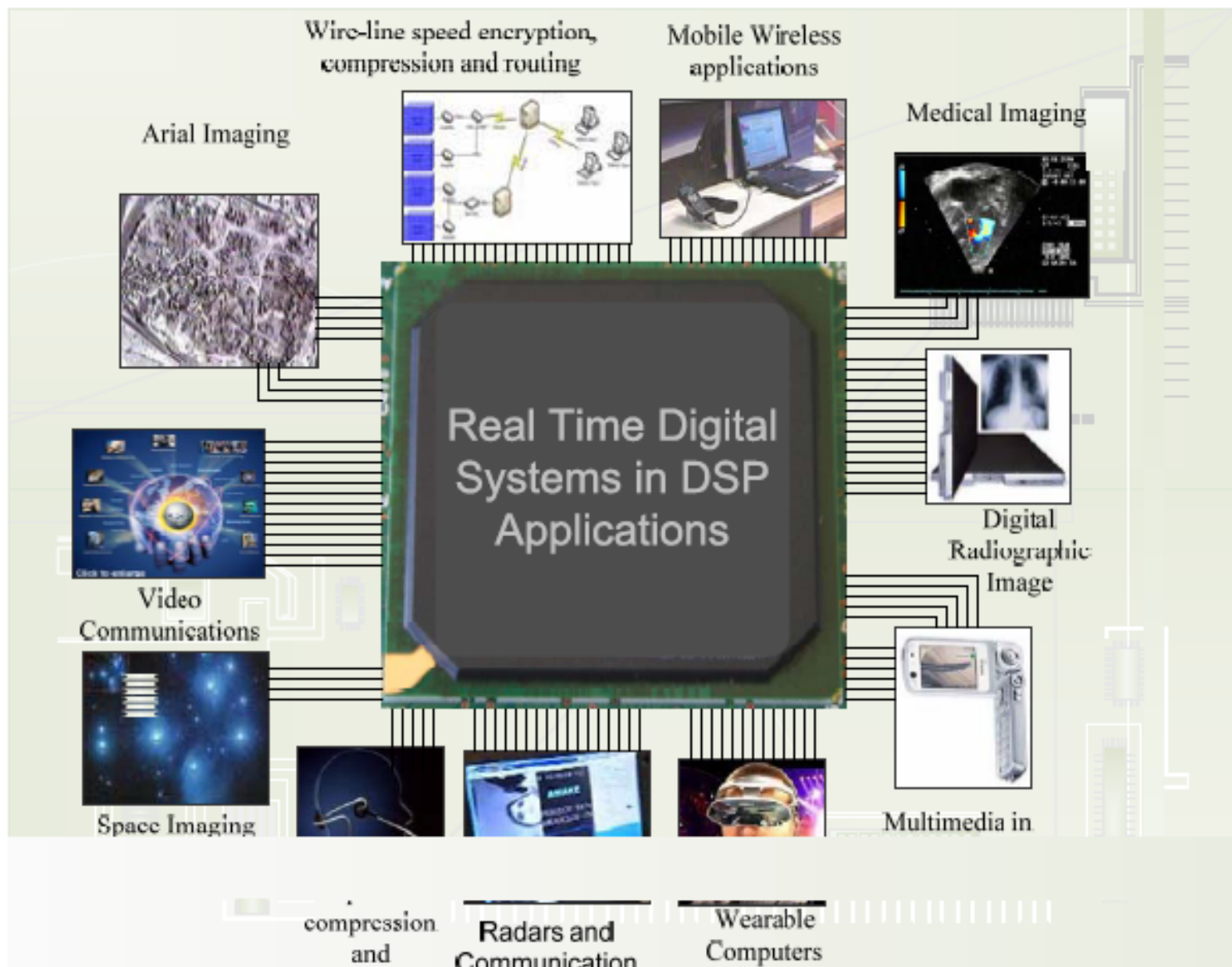
Digital System Design

Introduction

Intro

- Digital Design means
 - Mapping algorithms/applications in silicon
 - Applying transformations and tricks that results in optimal mapping in competing design space of
 - Area
 - Power dissipation
 - Performance
 - Testability
- VLSI has enabled solutions to intractable engineering problems
- VLSI has revolutionized the commercial market

Digital Systems in DSP Applications



Scope

- Digital System
 - Architecture
 - Technology (uC, uP, DSP, FPGA, ASIC)
 - Implementation
 - Fixed point arithmetic
 - Data-path
 - Combinational logic
 - Sequential logic
 - Memories: FIFOs, LIFOs, DPRAMs
 - Controller
 - FSM (Mealy, Moore), ASM, Micro-coded FSM

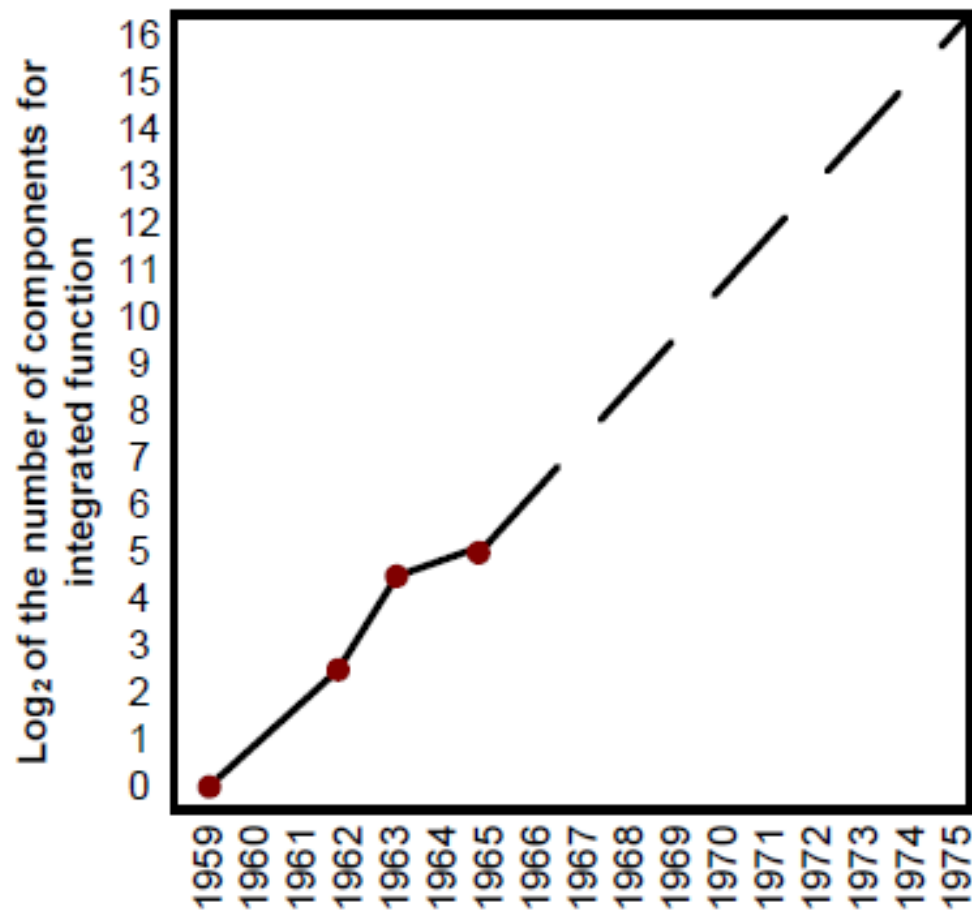
Digital Systems technologies & constraints

- Usually such a system consists of heterogeneous physical devices such as
 - Microcontrollers
 - Microprocessors
 - DSPs
 - FPGAs
- Choice depends on application
- Constraints?

Example

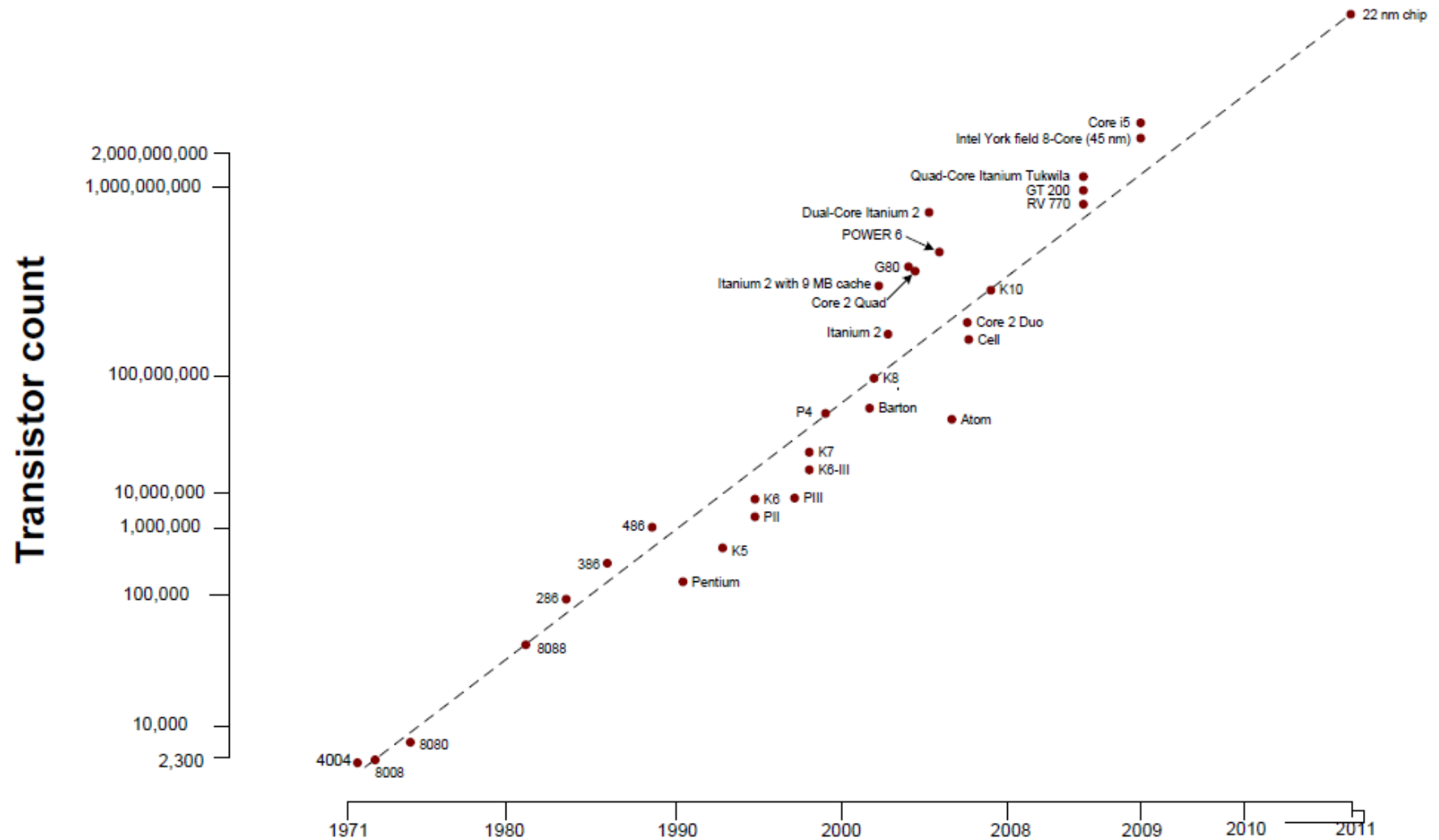
- Estimate processing requirement of speech
- Estimate processing requirement of video

Moore's Law



Electronics, April 19, 1965.

Intel continues to pursue Moore's Law



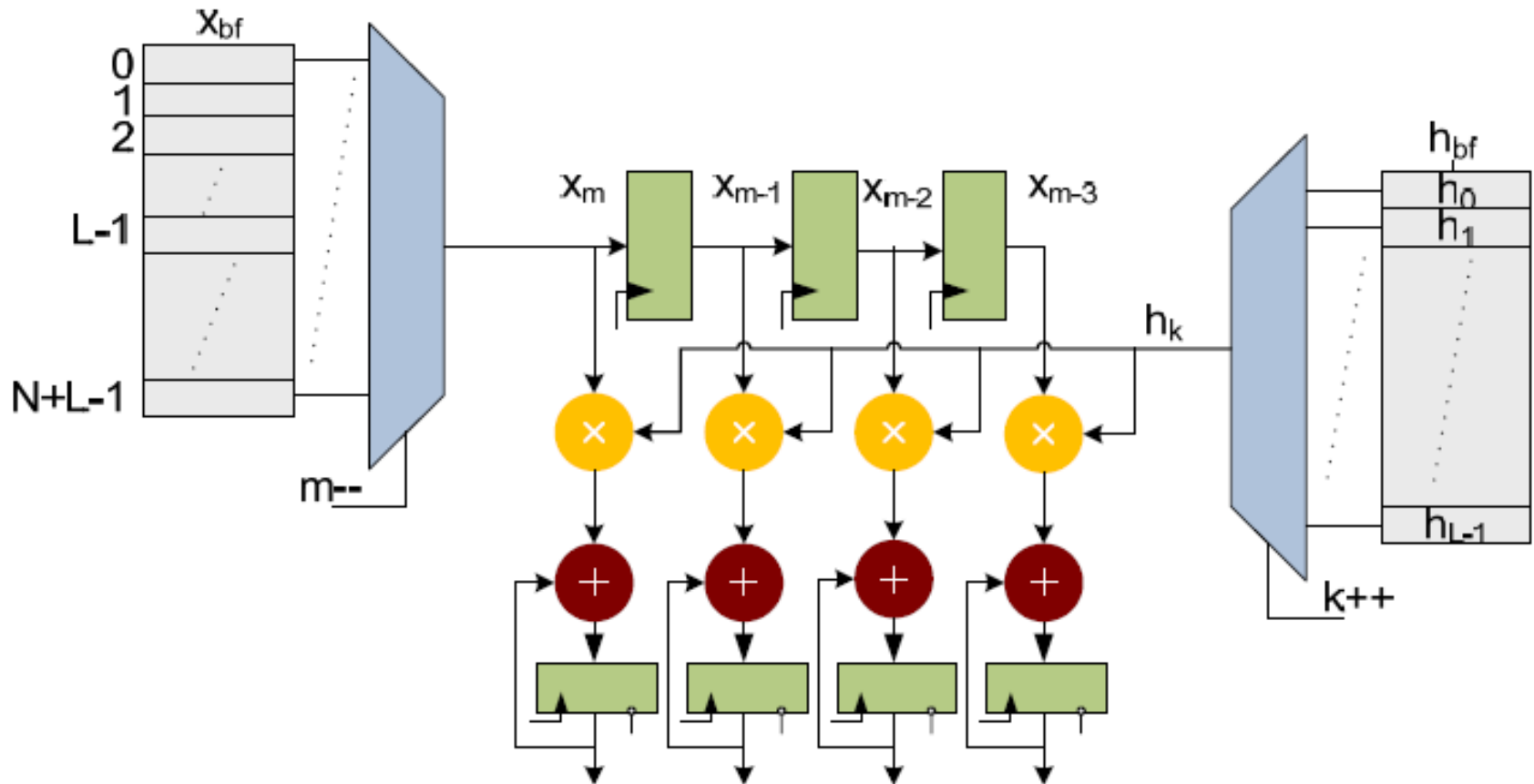
Intel tick-tock model

- https://en.wikipedia.org/wiki/Intel_Tick-Tock

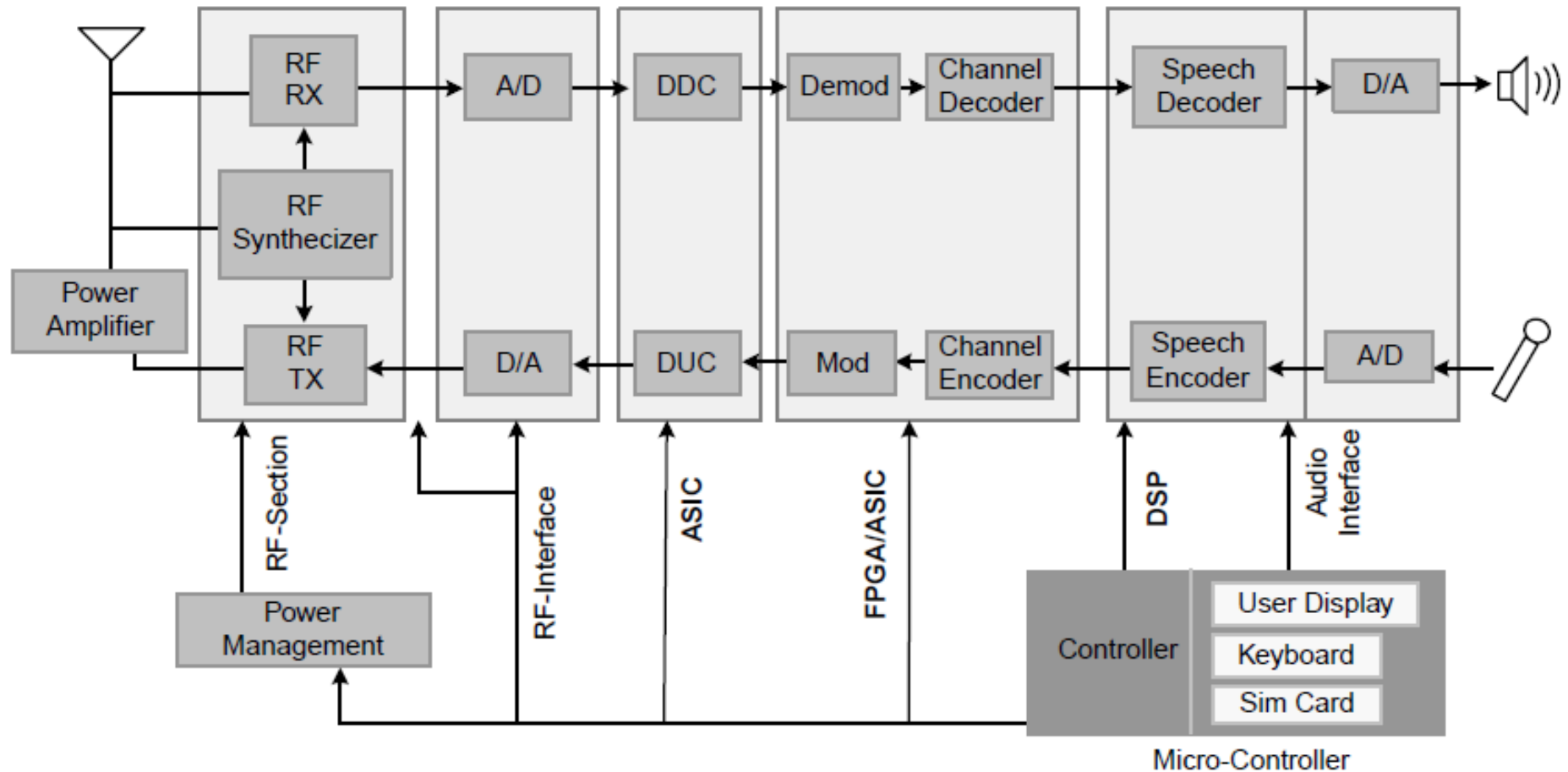
Algorithm implemented in C

```
#include <stdio.h >
#define N 12
#define L 8
// Test data and filter coefficients
short xbf[N+L-1]={1, 1, 2, 3, 4, 1, 2, 3, 1, 2, 1, 4, 5, -1, 2, 0, 1, -2, 3};
short hbf[L]={5, 9, -22, 11, 8, 21, 64, 18};
short ybf[N];
// The function performs block filtering operation
void BlkFilter(short *xptr, short *hptr, int len_x, int len_h, short *ybf)
{
    int n, k, m;
    for (n=0; n<len_x; n++)
    {
        sum = 0;
        for(k=0, m=n; k<len_h; k++, m--)
            sum += xptr[m]*hptr[k]; // MAC operation
        ybf[n] = sum;
    }
}
// Program to test BlkFilter and BlkFilterUnroll functions
void main(void)
{
    short *xptr, *hptr;
    xptr = &xbf[L-1]; // xbf has L-1 old samples
    hptr = hbf;
    BlkFilterUnroll (xptr, hptr, N, L, ybf);
    BlkFilter (xptr, hptr, N, L, ybf);
}
```

HW mapping of the code in previous slide



Digital Transceiver for a Voice Communication System

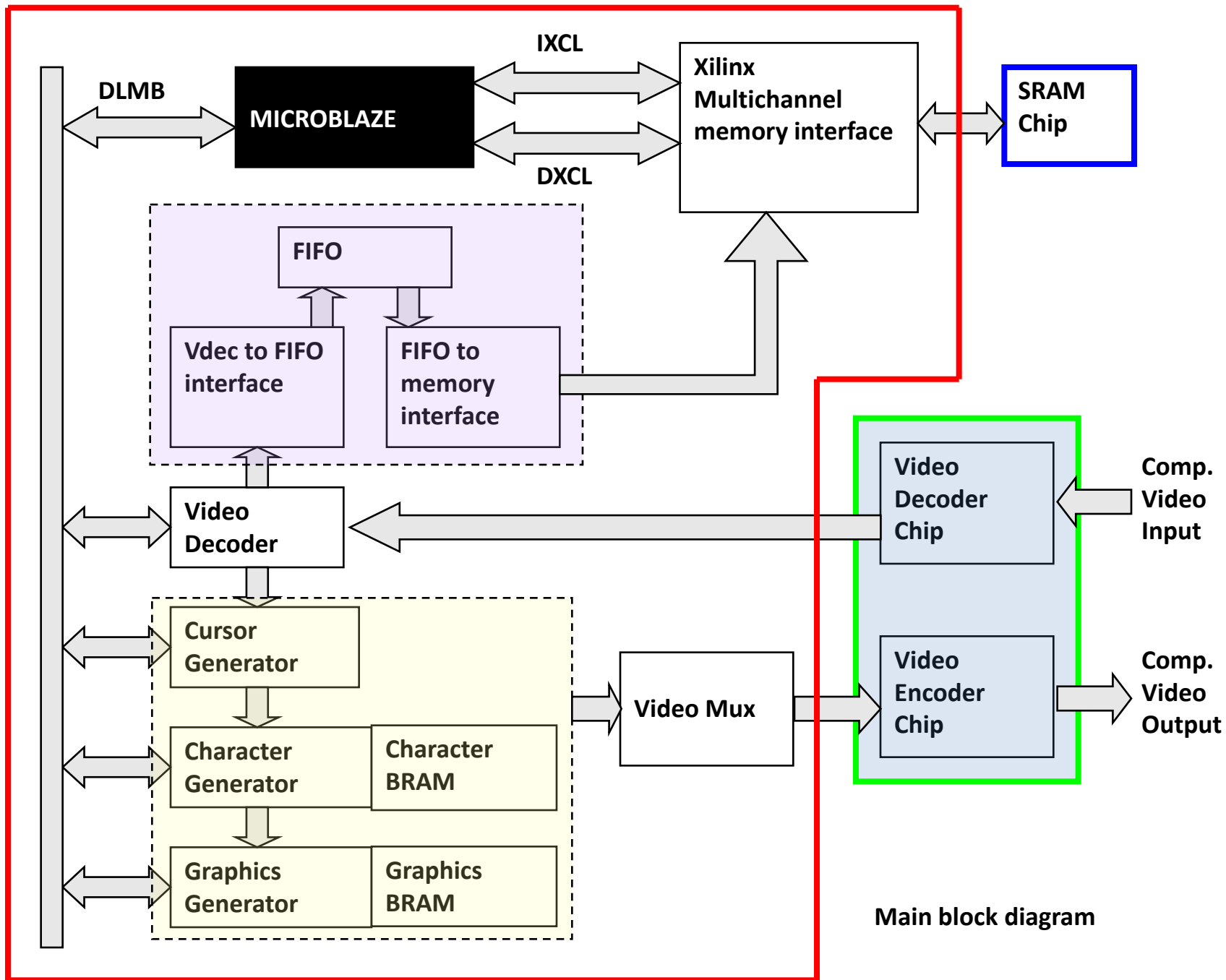


Algorithms distribution

- Code intensive algorithms
 - phone book management, keyboard interface
- Structured and computationally intensive algorithms
 - digital up and down conversion,
 - demodulation and synchronization loops
 - forward error correction (FEC)
- Code intensive and computationally intensive algorithms
 - Video/Speech compression

Design examples

Real-time surveillance system



Hardware



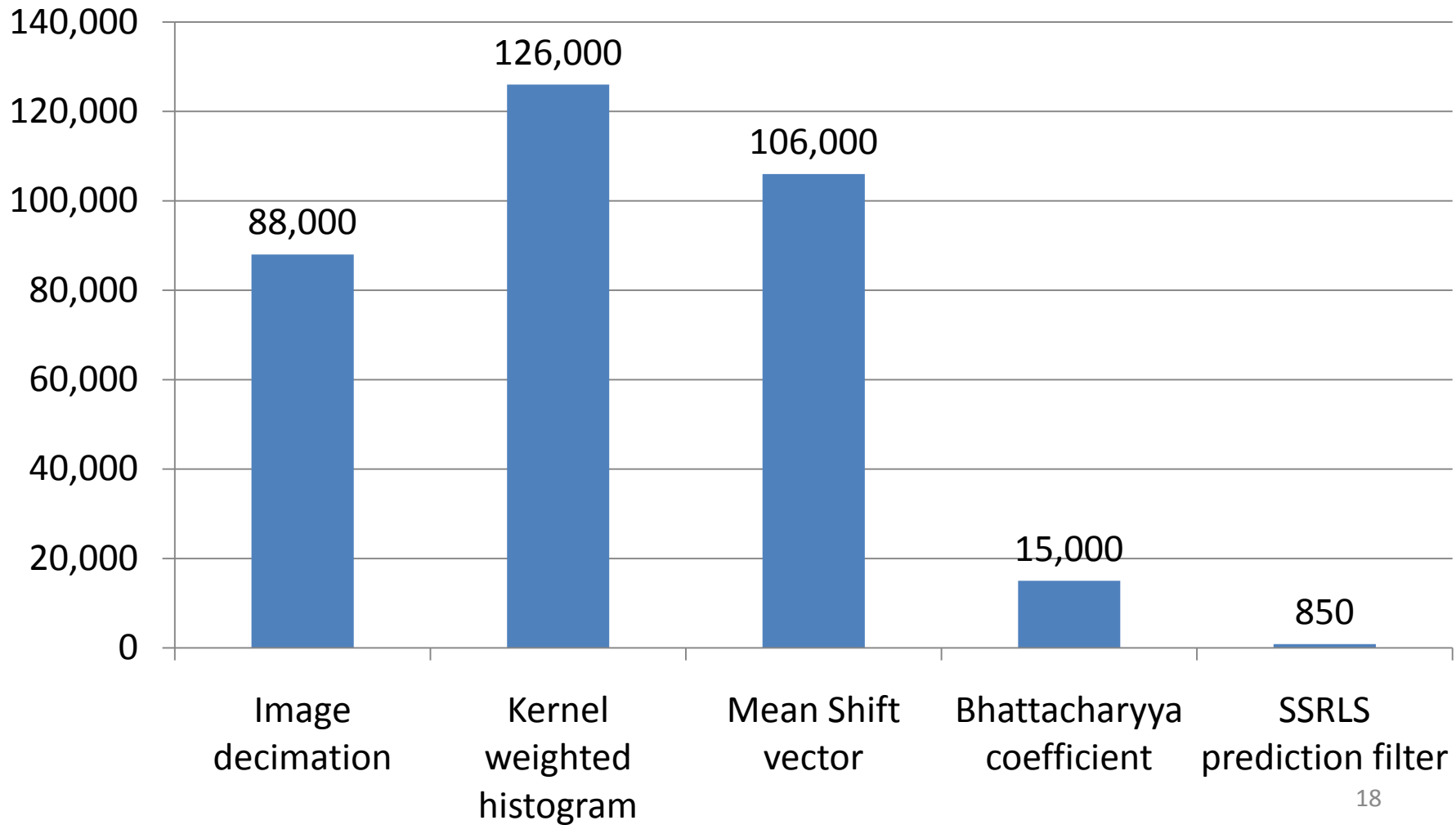
Design examples

Real-time surveillance system

Advanced version

Task analysis

Machine cycles



Hardware/software partitioning of Mean-Shift tracking algorithm

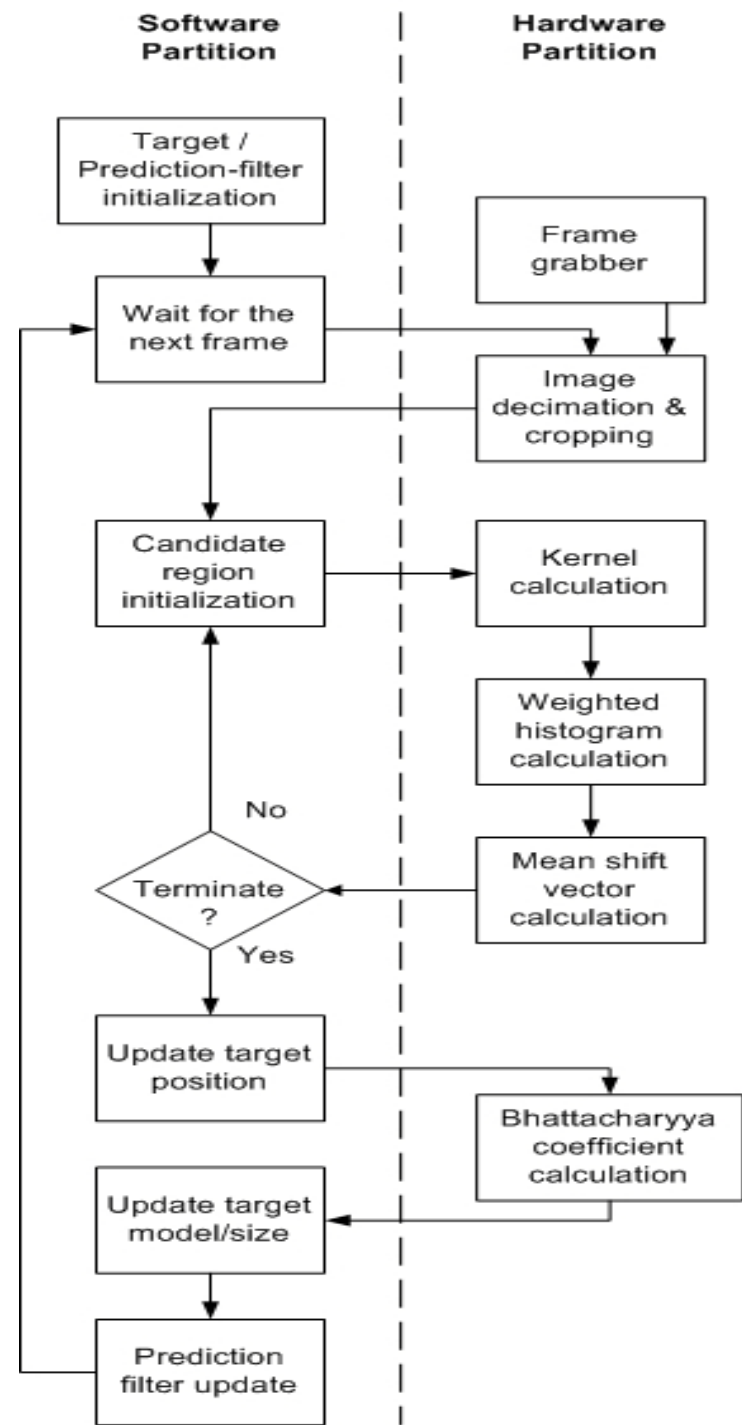
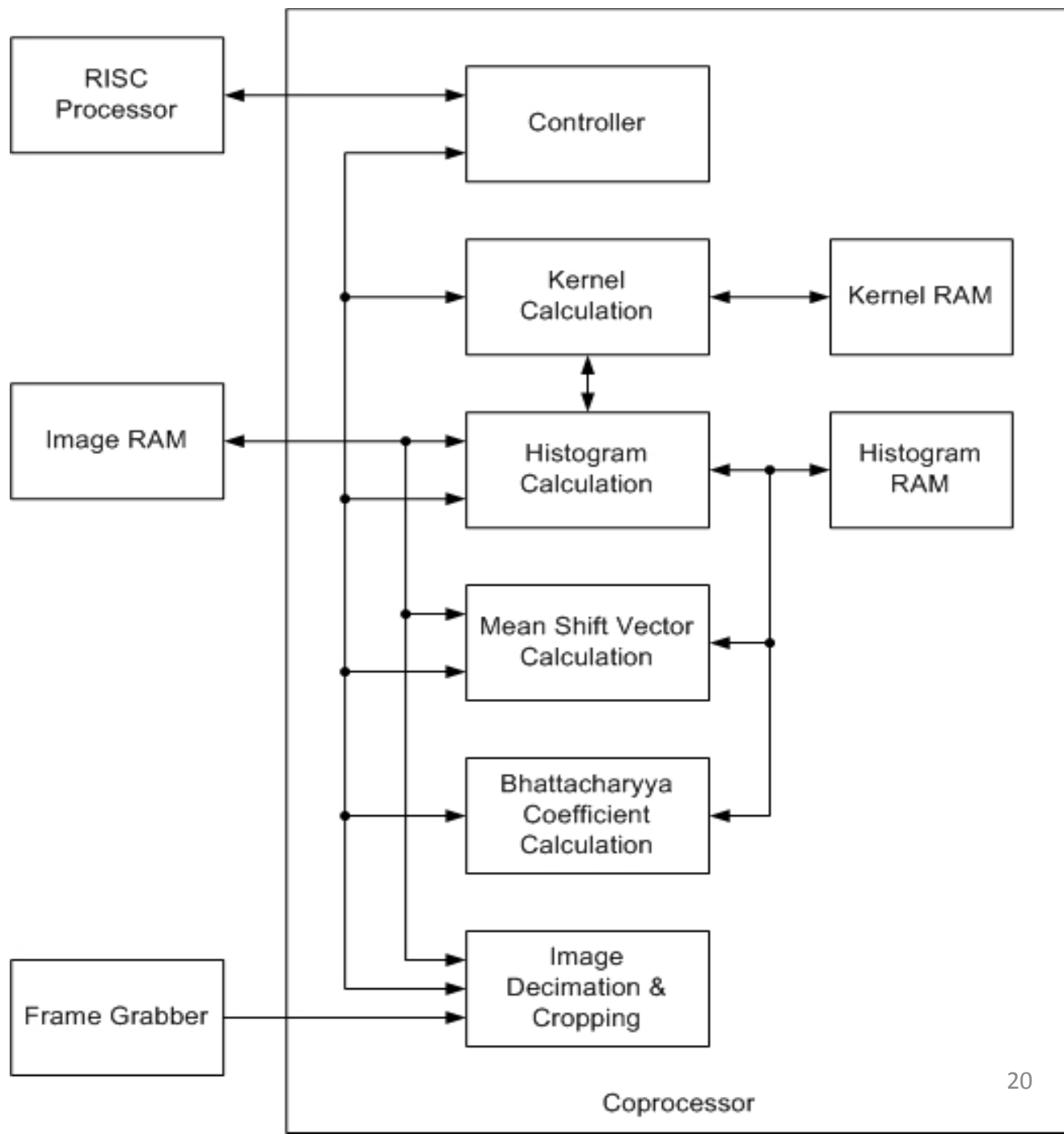
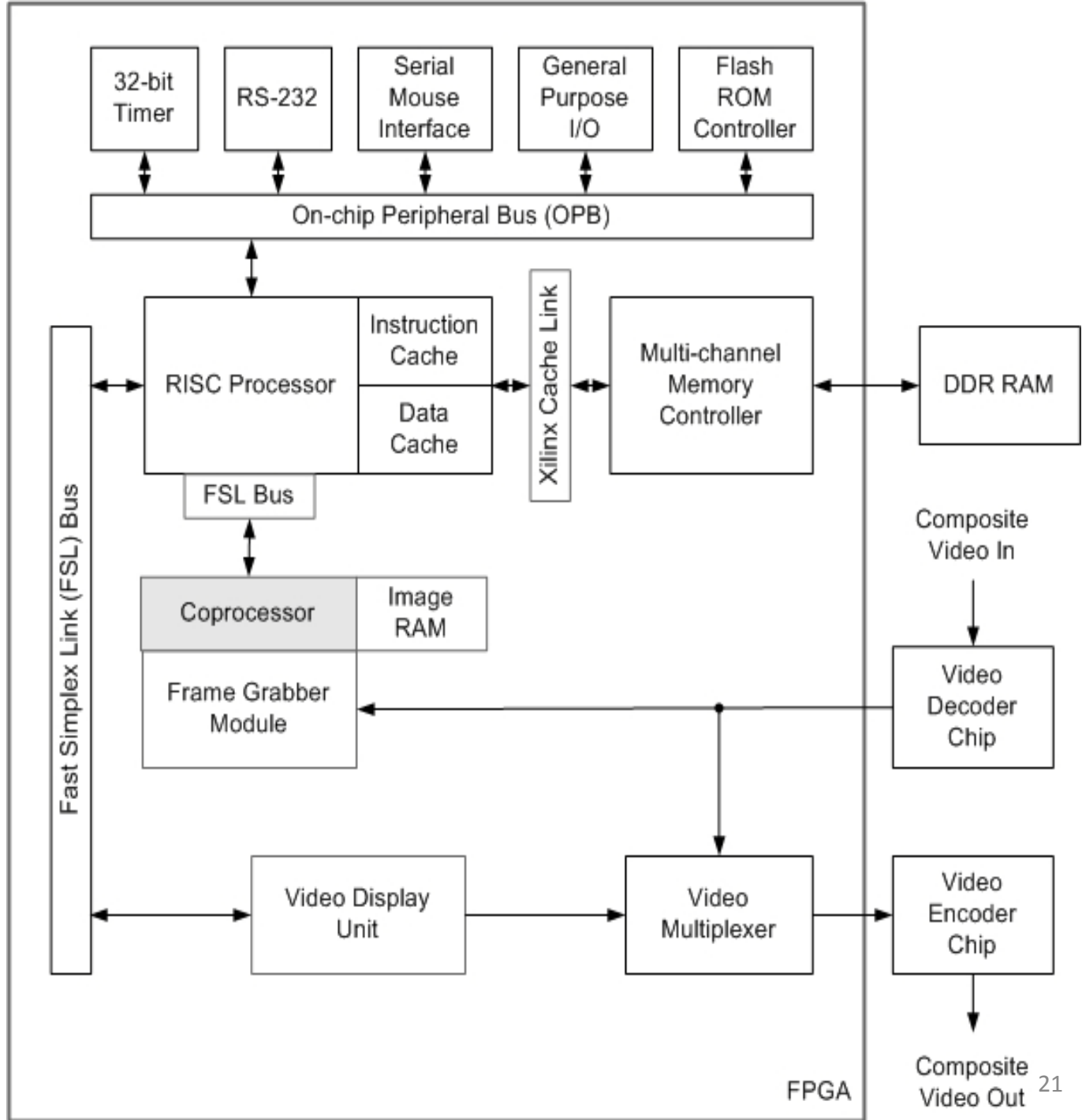


image coprocessor architecture (Mean-Shift tracker)



Proposed architecture hardware/software co-design



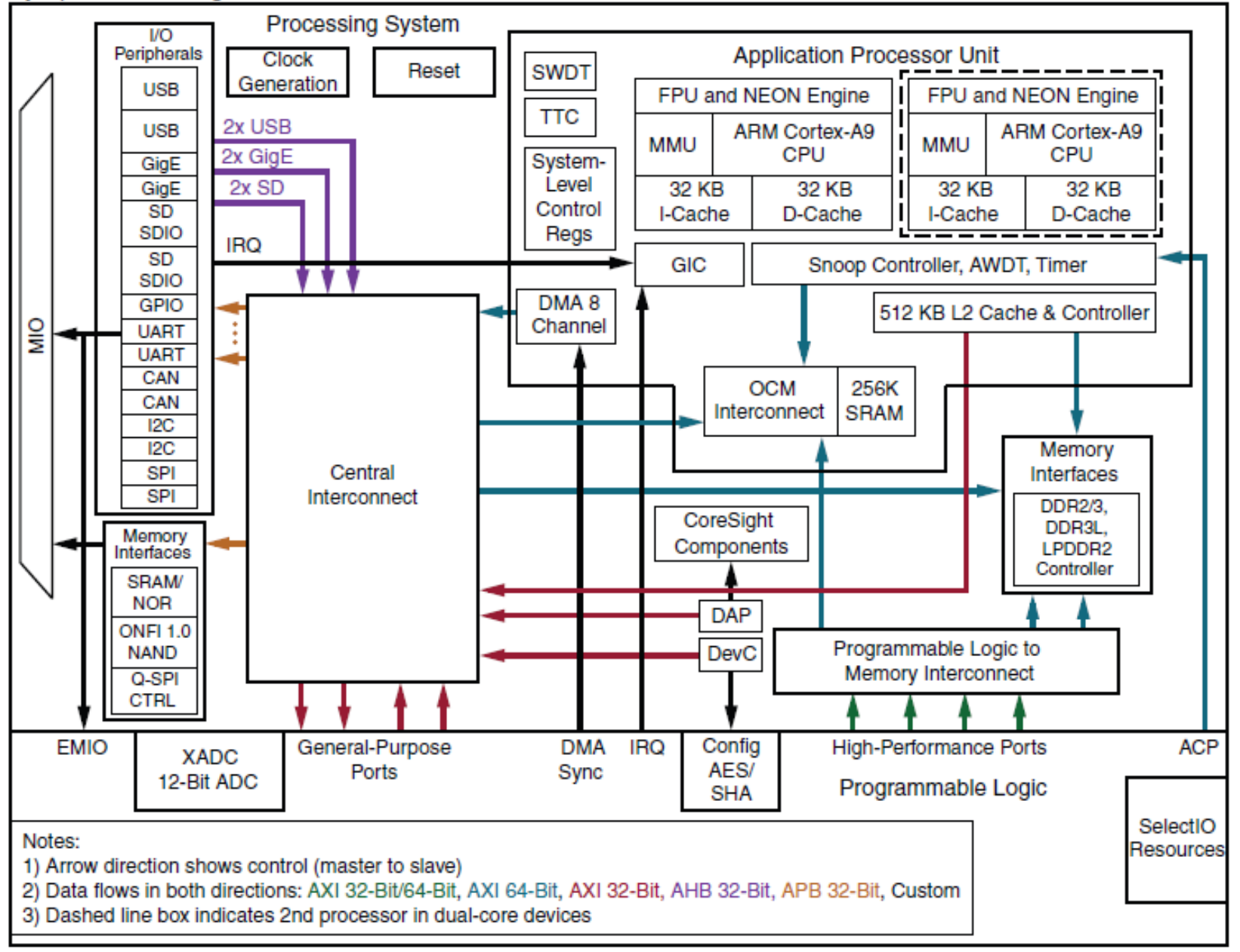
Performance comparison

- Software based Mean Shift tracker
 - Frame Rate = 10 fps
- Co-design based Mean Shift tracker
 - Frame Rate = 290 fps
 - **Acceleration = 29x**

Future directions

32-bit ARM processor embedded configurable logic

Zynq-7000 All Programmable SoC



Thank you